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RESEARCH ARTICLE

Design and Control of an Autonomous Bat-like Perching UAV

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Abstract
Perching allows small Unmanned Aerial Vehicles (UAVs) to maintain their altitude while significantly extending their flight duration and reducing noise. However, current research on flying habitats is poorly adapted to unstructured environments, and lacks autonomous capabilities, requiring ideal experimental environments and remote control by personnel. To solve these problems, in this paper, we propose a bat-like UAV perching mechanism by investigating the bat upside-down perching method, which realizes double self-locking in the perching state using the ratchet and four-link dead point mechanisms. Based on this perching mechanism, this study proposes a control strategy for UAVs to track targets and accomplish flight perching autonomously by combining a binocular camera, single-point LiDAR, and pressure sensors. Autonomous perching experiments were conducted for crossbar-type objects outdoors. The experimental results show that a multirotor UAV equipped with the perching mechanism and sensors can reliably achieve autonomous flight perching and re-flying off the target outdoors. The power consumption is reduced to 2.9% of the hovering state when perched on the target object.

Keywords Bio-inspired design · Perching mechanism · UAV · Autonomous flight

1 Introduction

UAVs can perform various airwork tasks in a wider three-dimensional space than mobile ground robots. Compared with fixed-wing or fluttering UAVs, multirotor UAVs have the advantages of a simple structure, low maintenance cost, high maneuverability, and can hover at a fixed point. With the maturity of sensors, control algorithms, microprocessors, and other technologies, multirotor UAVs have achieved valuable practical results in on-pass monitoring and management, environmental monitoring and modeling, emergency and disaster management, and infrastructure inspection [1–3].

However, multirotor UAVs generally have problems such as insufficient battery capacity and large operating noise. In actual tasks, UAVs are often required to perform long-term hidden operations, such as reconnaissance surveillance. Suppose a UAV can find and perch on natural targets, such as flying creatures. In that case, it can partially or completely stop the rotor motor while maintaining the height required for operation, significantly extending the flight duration and eliminating the noise generated by the rotors. The concept of flying perching has been proposed as a suitable solution.

Researchers worldwide have designed various perching methods. For example, non-mechanical methods include air pressure difference [4–7], chemical interaction [8–12], magnetic induction [13], and electrostatic induction. Liu et al. [5, 7] used a micro centrifugal pump on a drone to quickly discharge the air in the sealing cavity to generate pressure differences such that the drone could be adsorbed on a wall surface or ceiling. Kalantari et al. [11] installed a gripper made of dry adhesive on the front of a quadrotor UAV, and achieved adhesion by colliding the gripper with a smooth vertical surface. Garcia-Rubiales et al. [14] integrated modules such as electromagnets, metal plates, gas detectors, endoscopic cameras, and a six-rotor UAV to control electromagnets using pulse signals, allowing the system to dock on magnetic objects such as outdoor pipes to perform hazardous tasks such as flammable gas leak detection. Graule et al. [13] designed an insect-type satellite robot that used electrostatic adsorption to achieve repeated perching and detachment of almost any material. These innovative perching methods have considerably increased the operational range of UAVs. However, they are usually more demanding on the surface

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and environment and have less perching power, which greatly limits their application.

In contrast, the mechanical perching method is better adapted to outdoor unstructured environments because it mainly involves different types of perching mechanisms that interact with the perched object to generate perching forces. Various types of perching mechanisms have emerged, such as the micro spine type [15–19], modular [20], bird-like claws [21–30], and others [31–33]. Pope et al. [16] added a crawling mechanism with microspines to a quadrotor UAV, which could perch on a concrete wall and crawl like an insect. Doyle et al. [26], inspired by the songbird perching mode, designed a passive underdriven legged structure to achieve passive perching on tree branches using its self-weight. Bai et al. [21] studied the structure of a bird's foot and designed a perching mechanism that could actively change its configuration. They achieved perching on different objects outdoors by switching between different foot states. Roderick et al. [30] developed a bird-claw-like mechanical structure called SNAG. Its grasping action was driven by an artificial tendon and a tandem spring. When grasping an object, the impact force was converted into a greater grasping force by imitating the toe flexor and tendon locking mechanism of the peregrine falcon, while achieving self-locking to prevent rebounding and falling off. All these studies have made innovations and breakthroughs in the design of the bionic perching mechanism, which have significantly expanded the existing functions of UAVs and increased their endurance.

However, these flight perching methods are based on human manipulation of the UAV, which undoubtedly places high operational-level requirements on the UAV operator. At the same time, it is difficult for the operator to ensure a good view of the field operation to make operational decisions when the UAV performs remote monitoring tasks in the field. This limits the scope of application of UAV flight perching operations. Therefore, if a UAV can autonomously identify the perch target and complete the flight perch mission, it will undoubtedly improve the application value of flight perch. Researchers have made outstanding contributions to the domain of autonomous habitat attachment. Mao et al. [34] implemented autonomous perching of a small rotorcraft on inclined surfaces using machine vision and inertial sensors. However, the identification and localization of perched inclined surfaces relied on AprilTag, which was almost unattainable in real outdoor perching application scenarios. Popek et al. [28] proposed an autonomous flight perching system (AGRASP) that integrated robot sensing, 3D vision-based path planning, and perching operations on a quadrotor to achieve autonomous flight perching for cylindrical perching targets. However, its use of under-mounted robotic clamping made it less successful and less adaptable to unstructured perching environments because of its demanding requirements for the accuracy of localization of perching targets.

The main contributions of this paper to improve the success rate of machine-autonomous flight perching, and application scope are as follows.

1. We proposed an innovative bat-like autonomous perching multi-rotor UAV.
2. We designed a set of control strategies for a perching mechanism that can implement autonomous flight perching and re-flying.

The structure of this paper is as follows. Section 2 introduces the bionic and self-locking mechanisms of the perching mechanism. Section 3 introduces the environment sensing system integrated into a multi-rotor UAV designed for real-time localization of perching targets. Section 4 introduces the trajectory planning and control strategy for the autonomous approaching of targets to complete perching in an outdoor environment. Section 5 integrates the designed autonomous perching system into a quadrotor UAV to achieve autonomous flight perching and re-flying for typical targets in an outdoor scenario. Finally, the conclusions and summary of future work are presented in Section 6.

2 Bat-like Perching Mechanism Design

Flying creatures switch between perched and flying states aided by hand–eye coordination aided by hand–eye coordination. First, they identify the target for perching, such as branches, cables, and street lights. Then, they control their flight to approach the target until the gripper touches the target and grips it with force. The muscles receive force feedback from the perching target to stop flying and complete the perching action. As typical flying perching creatures, bats can perch upside down on tree branches to avoid predators and reduce energy consumption. At the same time, they can constantly observe and detect prey entering the surveillance area from their perching position, and quickly adjust themselves to complete the predation. The upside-down perching pattern of bats is due to a special Tendon Locking Mechanism (TLM, Fig. 1a) in the toe portion [35–37], which consists mainly of fibrocartilage cells (TX, Fig. 1a) located on the plantar surface of the long toe flexor tendon and proximal flexor tendon, and the synovial layer (FS, Fig. 1a) of the flexor tendon sheath. The tendon membrane of the toe deep flexor (FDP; Fig. 1a) of the bat is attached to the terminal phalanx (UP; Fig. 1a) at one end and directly to its upper body at the other end. This type of ratchet mechanism can pull the tendon tightly when the bat is hanging upside down, driving the class ratchet mechanism to engage and complete

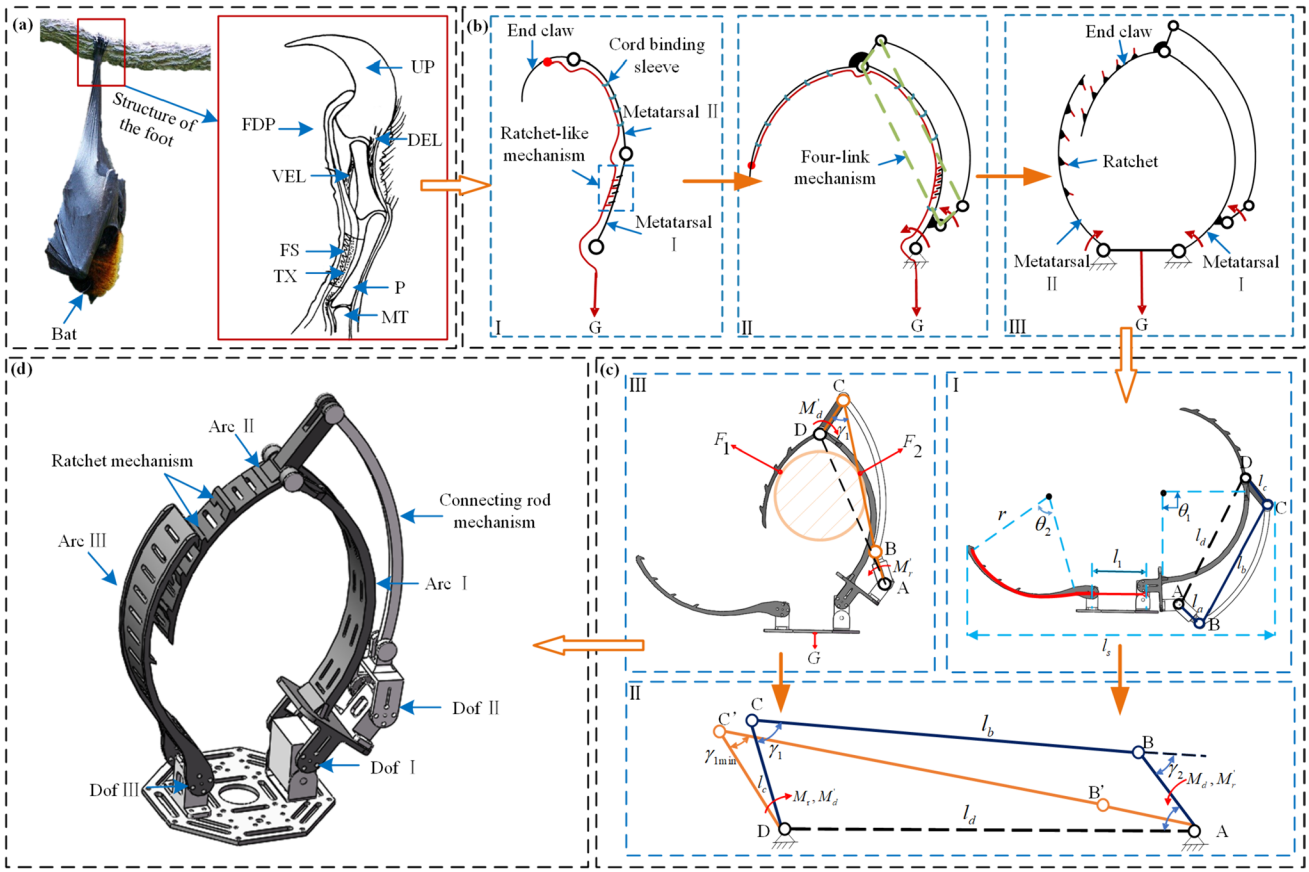


Fig. 1 **a** Bat perching mechanism, (*DEL* Dorsal Elastic Ligament, *FP* Flexor Process, *FS* Plicated Portion of Fexor Tndon Seath, *MT* Mtatarsal, *P* Poximal Phalanx, *TX* Txtured Patch on Plantar Surface of Flexor Tendon, *UP* Ugual (distal) Palanx, *VEL* Vntral Eastic Ligament.)[35]; **b** design of the self-locking mechanism and dimensional design of the perching mechanism, **c** dimensions of the perch attachment mechanism and force analysis of the four-link mechanism; **d** final design structure of the perch attachment mechanism

self-locking, thereby allowing the bat to hang for a long time by reducing muscle activity and energy consumption.

Unlike most birds, the bat's upside-down perching method is less obstructive to the field of view below. It uses gravity to stabilize its posture passively, making this perching especially ideal for flying robots. Inspired by the unique toe structure of bats, we simplified the structure, as shown in Fig. 1b, which has three main rotational degrees of freedom. The cord fixed at the end of the claw is connected to the main part of the UAV through a cord sleeve. When the UAV is perched and hung on the target, its gravity drives the cord downward to make the end of the claw lock the target automatically. At the same time, following the tendon self-locking mechanism of bats, a self-locking ratchet mechanism is designed on the surface of the cord and the surface of the metatarsal I. However, due to the small radius of the end claw, through this mechanism, it is difficult to adapt to a large-size target. Hence, we increased the radius of the arc of the end claw and simplified the three degrees of freedom to two degrees of freedom. To reduce the center of gravity of the mechanism, the drive of the top degree of freedom was moved to the bottom by a linkage mechanism, forming a sketch of the mechanism as shown in Fig. 1b II. However, considering the difficulty of making a cord with a ratchet-like mechanism, we removed the cord and increased the ratchet teeth on the back of the end claw, using another ratchet that can engage with it to form metatarsal II of the ratchet mechanism. Thus, the mechanism can not only achieve self-locking but also completely wrap around the perched target to prevent side-swiping caused by unilateral wrapping, as shown in Fig. 1b III.

Based on this bionic mechanism, we designed the perching mechanism shown in Fig. 1c, which contains three segments of three arcs wrapped around the perching target, a four-link mechanism, and a ratchet mechanism. We considered the axis distance of the multi-rotor platform (equipped with this perching mechanism) to be within 450 mm. To prevent excessive size from affecting the center of gravity of the UAV, we designed the length of the center plate of the UAV frame as $l_1 = 80$ mm, the radius of the three segments of the three arcs as $r = 128$ mm, the center angle of arc I $\theta_1 = 90°$, the center angle

of arcs II and III $\theta_2 = 60°$, the angle of the ratchet as 45°, and $l_s = 443$ mm when the mechanism is fully opened.

In the process of perching, the driving torque M_d generated by the motor acting on the active crank AB is transmitted to hinge C of the output member rocker CD in the form of a force through the connecting rod BC. This force drives it to overcome the working resistance moment M_r to work externally and realize the rotation of arc II, as shown in Fig. 1c II. γ_1 is the drive angle of the forward drive and γ_2 is the drive angle of the reverse drive. In the process of perching, a larger γ_1 can reduce the required motor driving torque to reduce energy consumption, whereas, in the closed state of the mechanism, a smaller γ_2 opens the mechanism with a larger resistance torque to improve reliability in the closed state. Based on the mechanical design experience, we set the minimum drive angle $\gamma_{1\min} \geq 45^{\circ}$ for the forward drive and $\gamma_{2\min} \leq 5^{\circ}$. for the reverse start. The length of the frame AD, $l_d = 213$ mm, was obtained from the dimensions of arc II and the motor. The final design dimensions were $l_a = 37$ mm, $l_b = 211$ mm, and $l_c = 55$ mm, considering the dimensions of the mechanism.

When the mechanism is perched on a cylindrical target with the ratchet not engaged, the final force is shown in Fig. 1c III. In the process of perching, the rudder at the Dof I is firstly controlled to keep the arc I and arc II upward. The rudder at the Dof II is controlled to rotate the arc II after contacting the perching target to wrap the target, and finally, the rudder at the Dof III is controlled to rotate the arc III to make the ratchet engagement self-locking. The motion process of disengaging from the perching is the exact opposite of the perching process. Due to the UAV's gravity G, the perched target generates a reverse pressure F_1, F_2 on the circular arc II, and this pressure acts on the connecting rod CD to generate the reverse driving moment M'_d. The resistance moment on the crank AB that must be overcome by this moment is M'_r. The corresponding four-rod mechanism is shown in Fig. 1c II, which $\gamma_{1\min}$ is approximately 45°, $\gamma_{2\min} = 0^{\circ}$ and the mechanism moves to the dead position. At this time, if the motor powers off, the reverse drive force distance M'_d cannot output the effective force to enable reverse rotation of the crank AB. Only the motor to the crank end of the reverse active torque can make the perching mechanism leave the target. Hence, even if there is no ratchet engagement, self-locking is possible, and the reliability of perching is improved. The perching mechanism was modeled in 3D according to the design dimensions, and the final results are shown in Fig. 1d.

3 Environmental Sensing System

To enable the UAV to perform the perching task autonomously, it should first track the perching target in real-time and obtain the relative position and attitude information. Therefore, we designed an adaptive real-time detection and tracking algorithm based on a priori target features, and the detection process is shown in Fig. 2.

First, the perched target points are artificially selected in the first frame of the image returned by the UAV. Then, the adjacent regions in the image with similar properties to the seeds, are merged using the region-growing algorithm [38] based on the color characteristics of the target points. The region is continuously expanded to extract the target. Based on the first extracted image, the perched target pixel degree, aspect, and external rectangular position are obtained as a priori information. The speed and acceleration of the UAV are not high during the flight perching task, and it is possible to control the changes in the target position under the pixel value coordinate system. Hence, the interference of the surrounding environment objects can be reduced by using the average pixel grayscale of the perching target obtained from the initial frame and the external rectangular position to perform image segmentation. After completing the preliminary image segmentation, the horizontal angle between the UAV and the target generally does not exceed 20°. The angle between the perched target and the horizontal axis of the pixel coordinate system within the image is small, and its longitudinal gradient features are obvious. Hence, the top hat operation can be used to remove the longitudinal element features, and the opening operation can be used to eliminate the parts of the image that are smaller than the structural elements but show details of these elements. Then, a closing operation is performed on the image to prevent small cracks or local color mutations on the surface of the perched target, from affecting the target recognition accuracy. Finally, the a priori features of the target aspect ratio are utilized to match all the rectangular boxes in the image. Subsequently, the a priori information needed for processing the next frame is updated based on the matching results. The recognition algorithm execution flowed in a complex indoor background with a horizontal tree trunk as the recognition target, which is shown in Fig. 2.

As outdoor Global Positioning System (GPS) positioning accuracy can only reach the meter level, it is difficult to achieve precise positioning of small perched targets. We used two single-point LiDARs placed directly in front of and below the UAV, as shown in Fig. 2. By performing Kalman filtering on the LiDAR information, we obtained relatively reliable information on the UAV flight altitude and the precise position of the target perched in front of it.

After tracking and positioning the UAV on the target, the UAV must be able to provide force feedback when it makes physical contact with the perching target, to achieve autonomous closure of the perching mechanism. Therefore, we arranged RP-L flexible thin-film pressure-sensitive sensors with elastic rubber bands and foam plates at the positions shown in Fig. 2 to simulate the muscles of bat

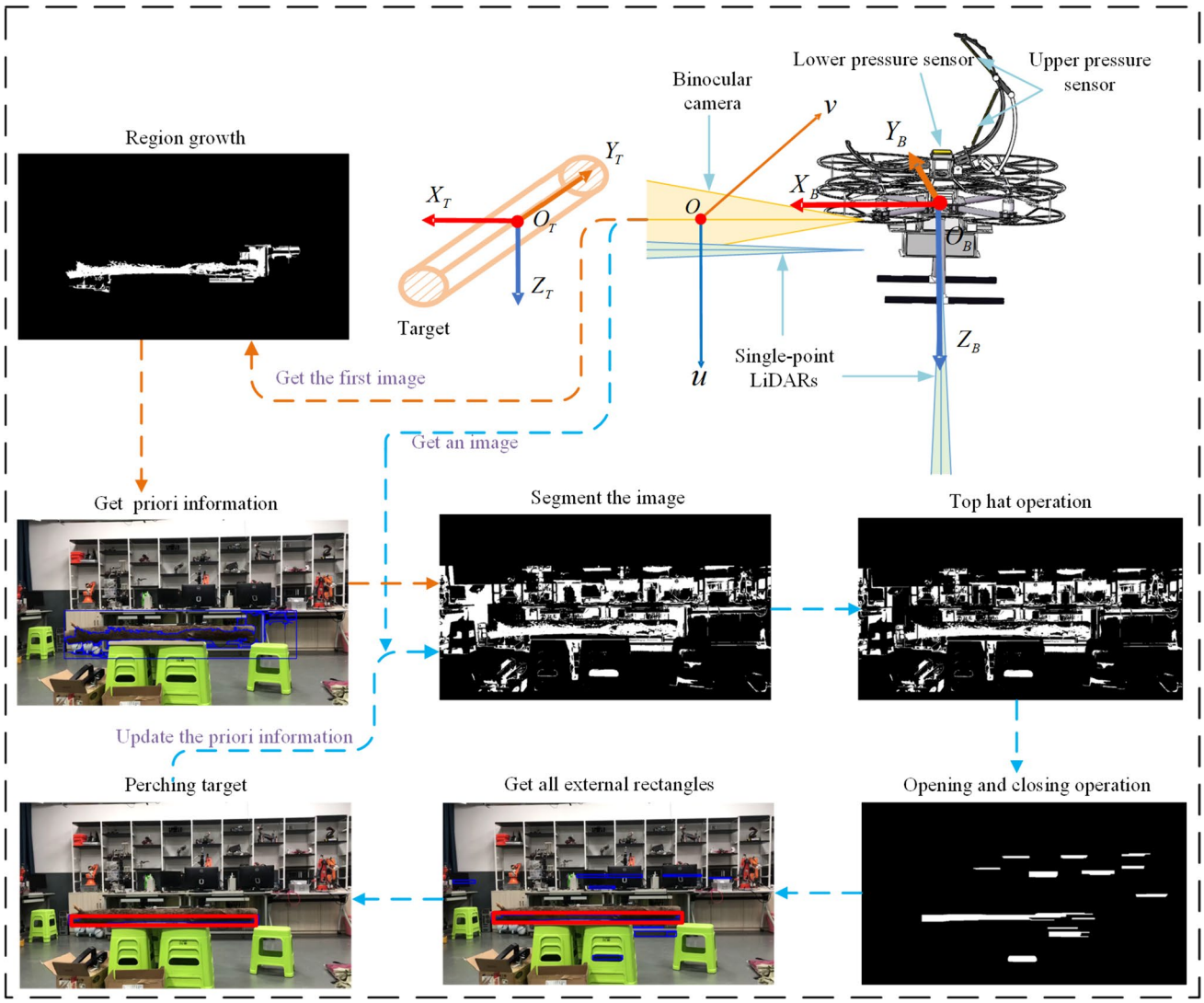


Fig. 2 Environment sensing system and the process of perched target extraction and tracking

claws to sense the contact pressure. These pressure sensors were mounted on the surface of the perching mechanism in contact with the perched target using rubber bands and cushioning foam. When perching, the upper-pressure sensor mounted on the arc would first contact the perched target, and when disengaging from the target and re-flying, the lower pressure sensor would contact the target due to the elevated altitude, and the pressure sensors would generate an electrical signal feedback when they physically contacted the perched target. The UAV controlled the corresponding movements of the upper and lower pressure sensors through the feedback from the upper and lower pressure sensors, respectively. When the pressure sensor made physical contact with the perched target, an electrical signal feedback was generated, and the UAV controlled the perching mechanism by the feedback from the upper and lower pressure sensors to perform the corresponding actions of perching and detaching, respectively.

4 Autonomous Perching Control Strategy

For a horizontal rod-like object as a perching target, with a binocular camera, it is difficult to obtain reliable depth information of the perching target in an outdoor long-range environment. To ensure the stability of autonomous UAV perching, we divided flight perching into three stages: center alignment, approaching the target, and perching completion. In the first stage, the UAV lifts off and hovers at a set altitude, and the coordinate system of the UAV and the perched target is established, as shown in Fig. 4. The coordinate system of the perched target is the fixed system $O_T - X_T Y_T Z_T$,

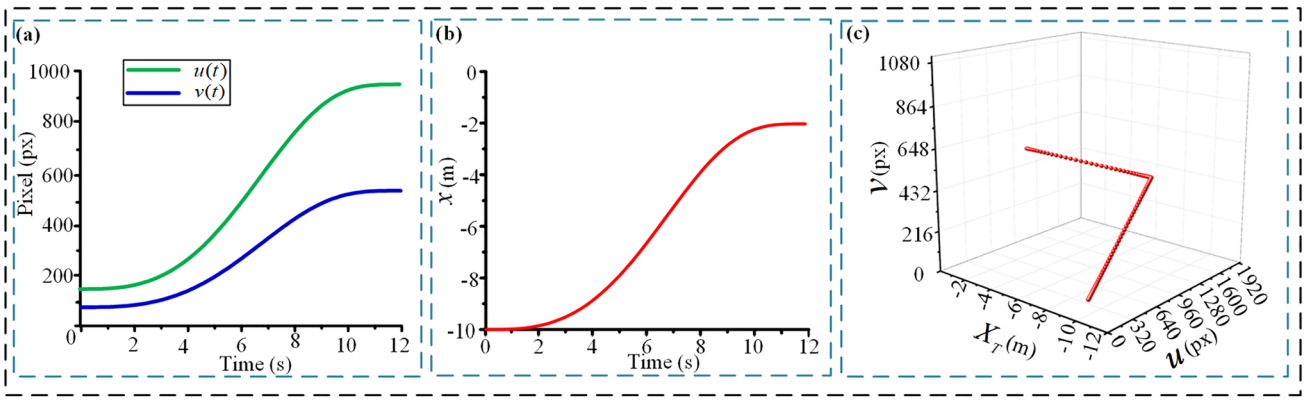


Fig. 3 **a** Simulation of the motion trajectory of the target center point in the first stage image coordinate system, **b** simulation of the UAV trajectory in the X_T direction in stage II; **c** simulation of the UAV trajectory in stages I and II

Y_T is towards the right along the length of the perched target, the Z_T axis is vertically downward, and the X_T axis is by the right-hand rule.

In stage I, the relative coordinates of the UAV and perched target have large deviations in the Y_T and Z_T axes, and the motion spacing of the Y_T and Z_T axes is mapped to the pixel coordinate system $o-uv$ of the binocular camera. The initial and termination velocities and accelerations of the UAV relative to the perching target are zero. Considering this, to make the perched target move to the center of the image under the motion of the UAV, we designed the Tau reference trajectory according to the internal Tau-j planning strategy proposed by Zhang et al. [39]. According to the TTC principle [40], the target coordinates in the u and v directions in stage I are defined as (w/2,h/2) within the $o-uv$ pixel coordinate system; w is the width of the image and h is the height of the image. The coordinates of the perched target center in the $o-uv$ coordinate system are $u(t)$ and $v(t)$. Then, the motion spacing $d_u d_v$ can be expressed as:

$$\begin{cases} d_u(t) = u(t) - \mathrm{w}/2 \\ d_v(t) = v(t) - \mathrm{h}/2 \end{cases} \tag{1}$$

The Tau of the motion spacing can be expressed as:

$$\begin{cases} \tau_u(t) = \frac{d_u(t)}{\dot{d}_u(t)} = \frac{u(t)-\mathrm{w}/2}{\dot{u}(t)} \\ \tau_v(t) = \frac{d_v(t)}{\dot{d}_v(t)} = \frac{v(t)-\mathrm{h}/2}{\dot{v}(t)} \end{cases} \tag{2}$$

The Tau of the motion spacing can be expressed according to the Tau-j planning strategy:

$$\begin{cases} \tau_u(t) = \mathrm{k_{uj}} \frac{1}{3}(t - \frac{t_d^3}{t^2}) \\ \tau_v(t) = \mathrm{k_{vj}} \frac{1}{3}(t - \frac{t_d^3}{t^2}) \end{cases} \tag{3}$$

Thus, the expressions for $u(t)$ and $v(t)$ are:

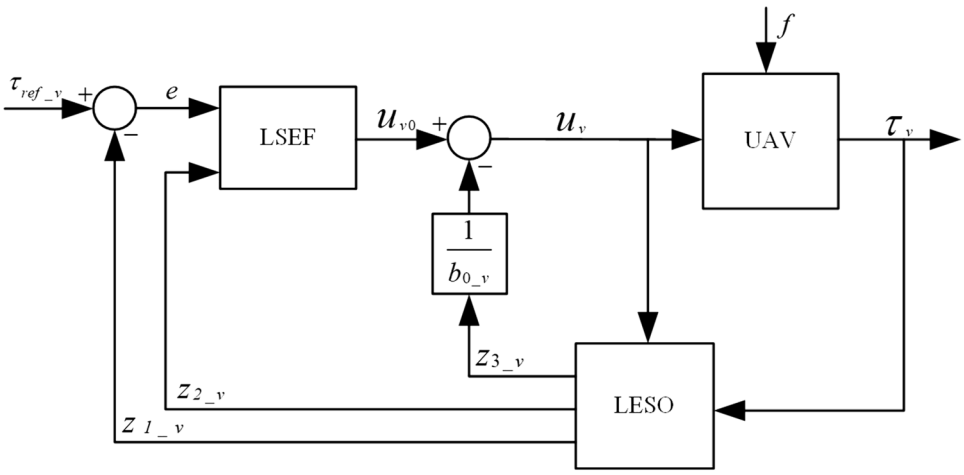


Fig. 4 *v*-direction LADRC trajectory tracking controller in pixel coordinate system

$$\begin{cases} u(t) = \frac{u_0 - w/2}{t_d^{\frac{3}{k_{uj}}}}(t_d^3 - t^3)^{\frac{1}{k_{uj}}} + \mathrm{w}/2 \\ v(t) = \frac{v_0 - h/2}{t_d^{\frac{3}{k_{vj}}}}(t_d^3 - t^3)^{\frac{1}{k_{vj}}} + \mathrm{h}/2 \end{cases} \tag{4}$$

u_0 and v_0 are the coordinates of the perched target center in the pixel coordinate system at the initial moment. Based on the image resolution, w is 1920 h is 1080, the closing time t_d is 12 s, u_0 and v_0 are 150 and 75, respectively, and $\mathrm{k_{uj}}$ and $\mathrm{k_{vj}}$ are set as 0.25. The simulated curves of $u(t)$ and $v(t)$, and the trajectory of the perched target in the pixel coordinate system are shown in Fig. 3a, where px is the pixel unit.

After completing alignment in the $\mathrm{Y_T}$ and $\mathrm{Z_T}$ directions, the UAV enters the second stage of autonomous perching. It acquires the current altitude through the lower single-point LiDAR and flies along the $\mathrm{X_T}$-axis at a fixed height to approach the perched target until the binocular camera samples the target at a distance of 2 m. The UAV also uses the Tau-j planning strategy to plan the Tau reference trajectory. At this time, the target state is 2 m behind the perching target, the coordinates of the UAV along the $\mathrm{X_T}$-axis are −2 m, and the current state is the relative distance between the UAV and the perching target along the $\mathrm{X_T}$-axis. Then, the motion spacing can be expressed as

$$d_x(t) = x(t) + 2 \tag{5}$$

The Tau of the motion spacing can be expressed as

$$\tau_x(t) = \frac{d_x(t)}{\dot{d}_x(t)} = \frac{x(t) + 2}{\dot{x}(t)} \tag{6}$$

Then, the trajectory of the UAV along the X_T direction can be expressed as

$$x(t) = \frac{x_0 + 2}{t_d^{\frac{3}{k_{xj}}}}(t_d^3 - t^3)^{\frac{1}{k_{xj}}} - 2 \tag{7}$$

Setting x_0 as −10 m, t_d as 12 s, and $\mathrm{k_{uj}}$ as 0.25, we simulated the curve of $x(t)$ as shown in Fig. 3b. The trajectory of the UAV in the pixel coordinate system $o - uv$ and $\mathrm{X_T}$ axis is shown in Fig. 3c.

To enable the UAV to accurately track the designed Tau trajectory under outdoor conditions such as wind speed changes and perturbations of the perching mechanism, to simplify the parameter adjustment process and perturbations of the perching mechanism, and to simplify the parameter adjustment process, we arranged a trajectory tracking controller on the upper layer of the servo layer based on Linear Active Disturbance Rejection Control (LADRC). This LADRC controller output is used as an input to the underlying flight control system to control the robot to follow the designed trajectory. We considered the trajectory tracking controller in the v direction under the pixel coordinate system as an example. Its controller structure is shown in Fig. 4.

In Fig. 4, τ_{ref_v} is the reference signal in the v direction, τ_v is the actual motion pitch measured by the binocular camera, and f is the total external perturbation to the UAV. LESO is a linear state observer that derives the estimated state variables z_{1_v} and z_{2_v} of the UAV and the total perturbation z_{3_v} of the system based on the outputs u_v and τ_v of the system. The linear state error feedback control rate LSEF is used to generate the feedback control quantity u_{v0}. The control algorithm of the LADRC trajectory tracking controller in the V-direction is as follows.

$$\begin{cases} \dot{z}_{1_v} = z_{2_v} - \mathrm{k_{1_v}}\left(z_{1_v} - \tau_v\right) \\ \dot{z}_{2_v} = z_{3_v} - \mathrm{k_{2_v}}\left(z_{1_v} - \tau_v\right) + \mathrm{b_v}\, u_v \\ \dot{z}_{3_v} = -\mathrm{k_{3_v}}\left(z_{1_v} - \tau_v\right) \\ u_{v0} = \mathrm{k_{p_v}}\left(\tau_{ref_v} - z_{1_v}\right) + \mathrm{k_{d_v}}\, z_{2_v} \\ u_v = u_{v0} - \dfrac{z_{3_v}}{\mathrm{b_{0_v}}} \end{cases} \tag{8}$$

In the above equation, $\mathrm{k_{1_v}}$, $\mathrm{k_{2_v}}$, $\mathrm{k_{3_v}}$ are linear state observer parameters, $\mathrm{k_{p_v}}$ and $\mathrm{k_{d_v}}$ are linear state error feedback rate parameters, $\mathrm{b_v}$ is the controller gain, and u_{v0} is an intermediate control quantity. Additionally, u_v is the UAV control quantity and is linearly mapped to the reference value of the UAV traverse angle ϕ_{ref}. The design parameters according to the bandwidth method are [41, 42]: $\mathrm{b_v} = \mathrm{b_{0_v}}$,$\mathrm{k_{1_v}} = 3\,\mathrm{w_{o_v}}$,$\mathrm{k_{2_v}} = 3\,\mathrm{w_{o_v}^2}$,$\mathrm{k_{3_v}} = \mathrm{w_{o_v}^3}$,$\mathrm{k_{p_v}} = \mathrm{w_{c_v}^2}$,$\mathrm{k_{d_v}} = 2\,\mathrm{w_{c_v}}$. The actual parameters to be adjusted are the controller compensation factor $\mathrm{b_{0_v}}$, the state bandwidth $\mathrm{w_{o_v}}$, and the controller bandwidth $\mathrm{w_{c_v}}$. Based on an actual UAV flight control system, the parameters can be adjusted to achieve trajectory-tracking control in a particular direction. Similarly, the LADRC controller with the same structure is arranged in the u and x directions, and the output is linearly mapped to the velocity reference value v_{ref} and pitch angle reference value θ_{ref} in the UAV altitude direction to achieve robust control of the UAV trajectory tracking.

When the distance between the UAV and the perched target is less than 2 m, it enters stage III. This stage requires physical interaction with the perched target to ensure that the pressure sensor is properly triggered upon contact. First, the UAV yaw angle is corrected to maintain the UAV camera parallel to the perched target on the axis. Considering the low depth-sampling accuracy of binocular cameras in complex outdoor environments, the UAV is allowed to rotate 30° left and right, and a simple PD controller is used to adjust the yaw angle of the UAV to the angle when the maximum length of the target perched in the image was obtained during the rotation. The angle is considered as the target angle when the UAV and target are parallel in the X-axis direction.

The UAV then makes three reciprocal movements of 0.5 m up and down, and the height data of the smallest distance measured by the forward LiDAR is regarded as the precise height of the perched target. The flight height of the UAV is adjusted such that the target is located in the gripping range of the mechanism. Finally, the pitch angle of the UAV is adjusted to approach the target until the surface pressure sensor is triggered to close the perched mechanism. Rotors are stopped to achieve successful perching.

After completing the perching task, the UAV starts the motor, increases the throttle amount to raise the altitude, and triggers the lower pressure sensor by contacting the perching target. At this point, the UAV maintains the altitude, opens the perching device, and then returns to flight mode to achieve autonomous re-flight.

5 Experiment

We created a perching mechanism by processing the resin material using light-curing and integrating it into a quadrotor UAV, as shown in Fig. 5a. The Jetson NX was selected as the upper layer controller, which communicates with the flight controller and binocular camera via its UART interface, and processes the data acquisition from single-point LiDARs via its IIC interface while running the task-related algorithms of target extraction and tracking, trajectory planning, and Kalman filtering. The DJI A3 flight controller was selected as the lower layer control system to control the robot flight, perch attachment mechanism, and signal acquisition of pressure sensors. The software was operated on the ROS system, which divided the sensor data processing, UAV flight control, and perching mechanism control into multiple nodes and carried out node communication through the topic mode, the structure of which is shown in Fig. 5b.

We conducted an autonomous perching experiment using the crossbar on the soccer goal frame of the playground as the perching target and placed the UAV in front of the soccer goal frame diagonally as shown in Fig. 6 stage I. A part of the target was in the UAV camera frame. After remotely selecting the crossbar as the perch object, an autonomous perch program was run. At the end of stage I, the perch target was already in the center of the image and the UAV adjusted its position to face the crossbar.

We recorded and plotted the motion spacing $d_u(t)$ in the u direction, velocity $\dot{u}$ and acceleration $\ddot{u}$ in the u direction, and the actual Tau trajectory using Jetson NX for stage I in the pixel coordinate system during the three experiments. The results are shown in Fig. 7a. We experimentally tuned the parameters of the designed LADRC controller, with the parameter values as described earlier: $b_{0_v} = 0.5$, $w_{0_v} = 400$, $w_{c_v} = 100$. It was evident that using the LADRC controller enabled precise tracking of the designed Tau theoretical curve. The motion spacing could be closed smoothly within the specified time, and the UAV motion process was smooth and could reach the set target. The velocity and acceleration of the pixel coordinate changed in the U-direction fluctuates somewhat but conforms to the trend of the theoretical

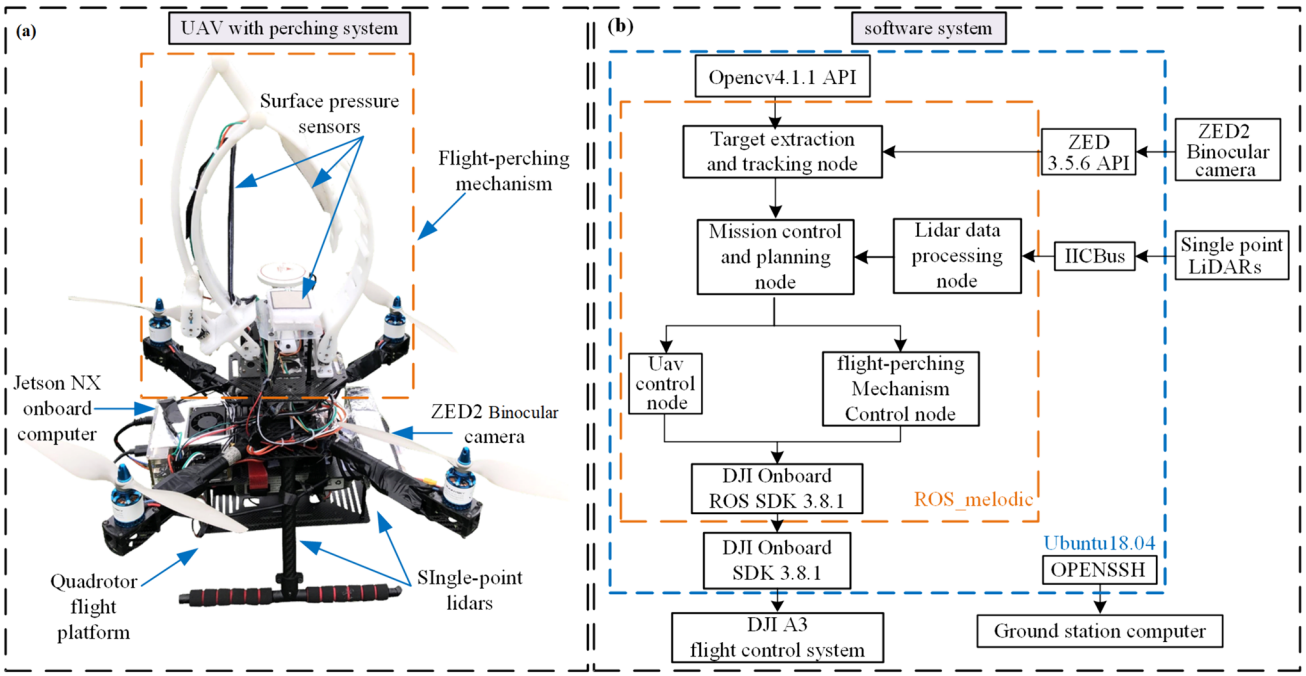


Fig. 5 **a** UAV after integration of autonomous perching system **b** robot software system

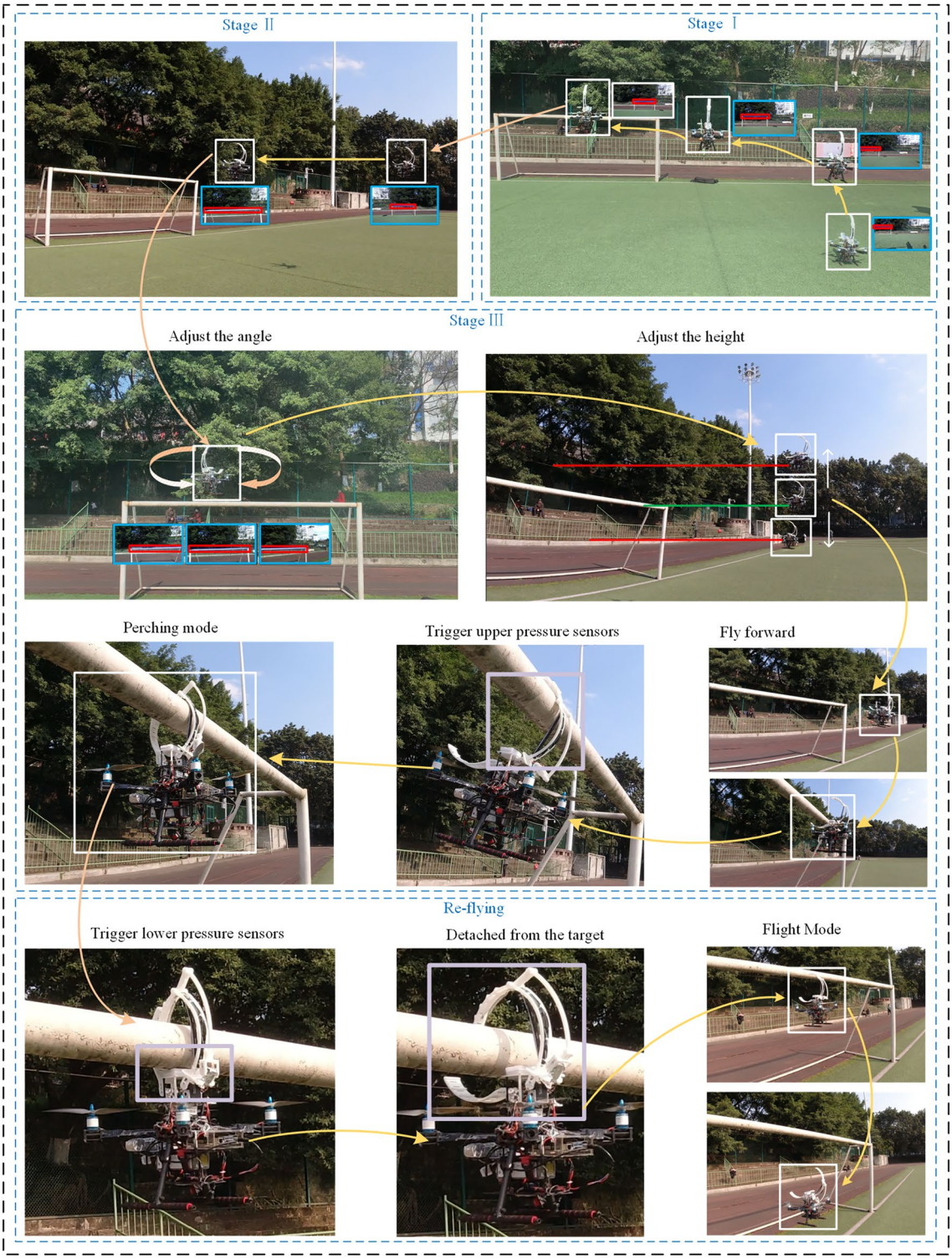


Fig. 6 Experiment on autonomous flight perch and re-flying of a UAV

curve, and the initial velocity, initial acceleration, termination velocity, and termination acceleration were all in line with the theoretical design close to zero. In stage II, the UAV moved forward with a fixed pitch angle along the X -direction with the help of a downward single-point LiDAR while maintaining the current altitude. The relative distance between the robot and the perching target was obtained in real-time using a binocular camera, as shown in Fig. 6 Stage

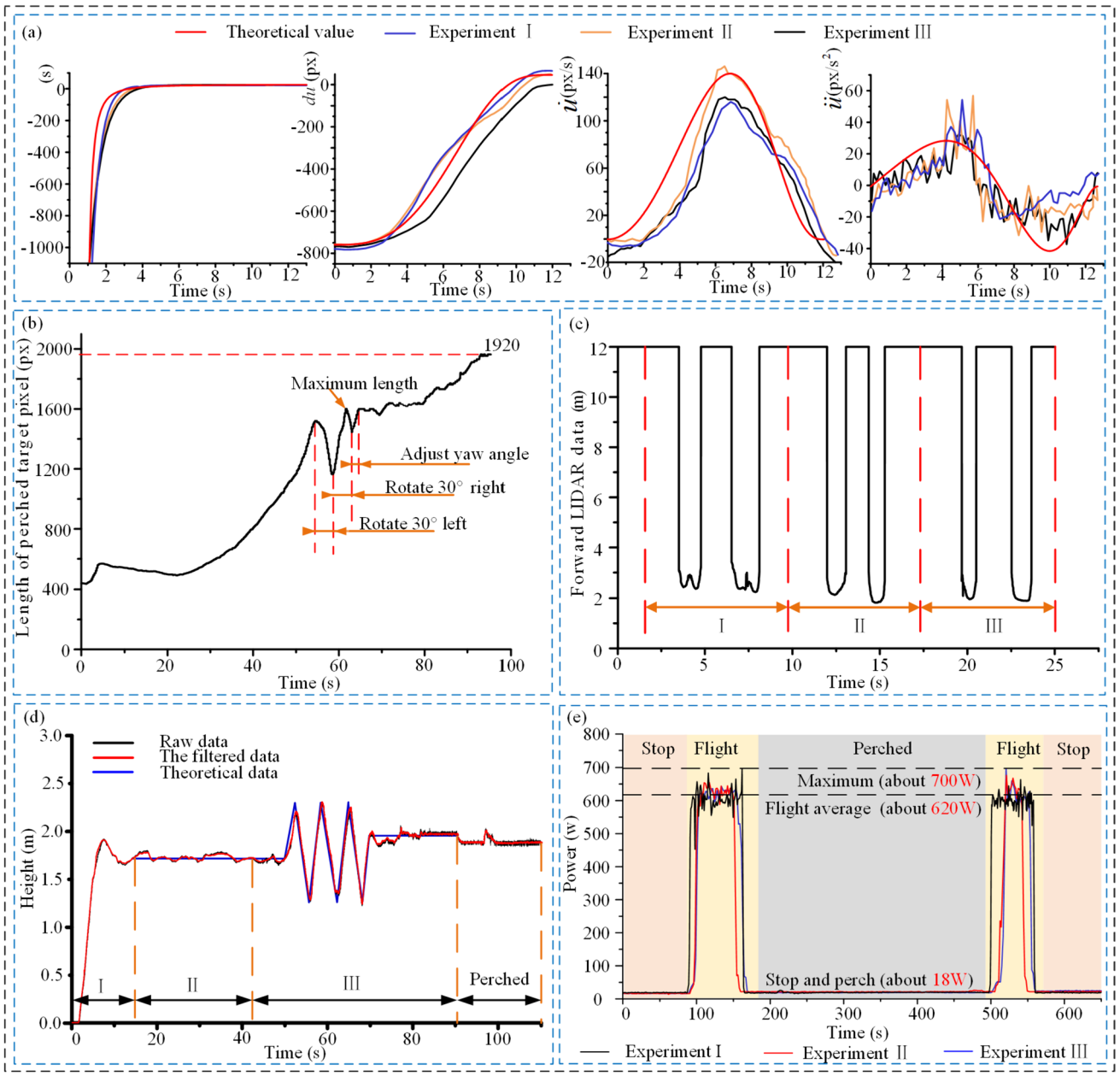


Fig. 7 **a** Tau-j theoretical curve and experimental curve, **b** pixel length of the target in stage III, **c** forward LiDAR distance data in stage III, **d** height data of the autonomous perching process of the UAV, **e** power consumption curve during the flight perch of the UAV

II. The movement stopped when the distance decreased to 2 m.

In stage III, the UAV adjusted the yaw angle by rotating 30° left and right in front of the perched target. The maximum length of the perched target in the image was used to achieve the opening of the mechanism to align the target. Then, up and down reciprocal motion was performed to determine the precise height of the perched target through the single-point LiDAR at two locations. The height-adjustment process is shown in Fig. 6 stage III. In the third detection stage, the lengths of the target pixels during the adjustment process were shown in Fig. 7b, indicating that there was a significant change in the length of the target during the rotation of the UAV, effectively aiding the UAV in aligning the perching target. The forward LiDAR detection distance was shown in Fig. 7c, demonstrating that the LiDAR data accurately fed back the forward distance during the up and down movement of the UAV when aligning the perched target, and at the same time, it helped the UAV determine the accurate height information of the target.

After completing the precise positioning of the target, the UAV adjusts its altitude to fly forward, when the altitude

of the perching target is already within the opening of the perching mechanism. The UAV continues to fly forward, and the mechanism closes after the perching target triggers the pressure sensor. The perching mechanism is then closed, the UAV rotors stop, and the UAV switches to perching mode as shown in Fig. 6 Stage III. When the UAV is required to switch from the perched mode to re-flying, it can be detached autonomously by triggering a lower pressure sensor, as shown in Fig. 6.

Figure. 7d shows the height data recorded by the downward single-point LiDAR during the experiment and after Kalman filtering throughout the autonomous perching experiment. It is evident that the introduction of the lower single-point LiDAR significantly improves the positioning accuracy by controlling the UAV height-fixing flight control between ± 0.15 m in Stage II and Stage III.

In addition, the power consumption of the entire aircraft was measured by connecting a curved power meter in series with the UAV power input. The power curve of the drone is presented in Fig. 7e. The quadcopter UAV equipped with the autonomous flight perching device consumes approximately 620 W of power during the takeoff and hovering phases and approximately 18 W when perched at a certain altitude, which is the same as when it is stationary on the ground. Hence, the power consumption is reduced to 2.9% of that in the hover mode while maintaining altitude.

6 Conclusions and Future Work

In this study, we designed a bat-claw-like perching mechanism that can be mounted on a multirotor UAV. We also designed an autonomous perching control strategy for UAVs based on it. Our experimental results show that integrating this device with a multirotor UAV makes it possible to achieve autonomous perching for typical crossbar-type targets in outdoor environments with significantly reduced power consumption. These results demonstrate the reliability of the designed autonomous perching solution, which significantly improves the application scope and practicality of flying perches.

Our future work will focus on the following two key areas. First, we will introduce machine learning and other methods to optimize the autonomous recognition and selection of perching targets to further improve the autonomy of perching. In addition, the current flying perching robots can only perch on typical static targets in the environment. If these can also perch on dynamic targets such as moving vehicles, their application scope will be further expanded. To achieve this goal, we need to further optimize the current perching mechanism so that it can adapt to more perching targets, such as flat or vertical poles. In addition, further research is required on dynamic trajectory planning and trajectory tracking control of UAVs.

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Data Availability All data generated or analysed during this study are included in this published article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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